higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

CAPENTRY AND ROOFWORK
NQF LEVEL 2

12 MARCH 2019

This marking guideline consists of 5 pages.

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QUESTION 1

1.1 Circular saw or jigsaw. (1)

1.2
- Tape measure
- Plumb bob
- Roofing square
- Electric drill
- Claw hammer
- Chalk line
- Spirit level (Any 5 × 1) (5)

1.3 A plumb bob consists of a length of chord with one end attached to a pointed metal weight.✓✓

Usage:
- Drive a nail about halfway in, horizontally, into the apex of the truss.
- Attach the end of the chord to the outer end of this nail.
- Allow the weighted end to hang free down the length of the king post.
- Wait until the weight comes to a rest.
- Measure the distance between the chord and the apex.
- Measure the distance between the chord and the tie-beam.
- If the two measurements taken are equal, the truss is plumb.
- If the measurements differ, the truss is not plumb and must be shifted until both measurements are the same.

Purpose:
A plumb bob is used to ensure that roof trusses are perfectly vertical.✓ (11)

1.4 A sliding bevel consists of a handle and an adjustable, slotted blade.✓
It can be used to measure✓ and transfer✓ angles. (3)
[20]

QUESTION 2

2.1 A stone axe. (1)

2.2 To cover✓ and protect✓ the interior of a building and its occupants. (2)

2.3
- Dead loads are permanent✓ loads on the roof such as air conditioning✓ and water tanks.✓
- Live loads are temporary✓ loads on the roof such as wind✓ and rain.✓ (6)

2.4
- Mono-pitch roof✓ – roof has only one slope.✓
- Hip roof✓ – all the sides of the roof end in a slope.✓
- Flat roof✓ – roof has a slope of less than 6 degrees.✓
- Conical roof✓ – round roof found on rondavels.✓
- Gable roof✓ – both ends of the roof are vertical.✓ (10)

2.5
- Rationally designed roofs are designed by engineers✓ using scientific calculations.✓
- Deemed-to-satisfy roofs are developed by craftsmen✓ over time, based on observation and experience.✓ (4)

2.6
- Spacing trusses correctly for the roof covering.
- Using roof anchors effectively.
- Using the right of wall plates for different roof spans.
- The use of tilting battens.
- Using different types of hurricane clip.
- Installing battens and purlins correctly.
- How to sling and lift a truss safely.
- Transporting, handling and storing and trusses. (Any 4 × 1) (4)

2.7 The wall plate spreads the weight of the whole roof evenly.
The wall plate is fixed flush with the inner skin on top of the external walls. (2)

2.8
2.8.1 38 mm × 76 mm
2.8.2 38 mm × 114 mm
2.8.3 38 mm × 152 mm
(3 × 1) (3)

2.9 200 mm. (1)

2.10 Battens are 38 mm × 38 mm✓ lengths of timber used to support roof tiles.✓
Purlins are 50 mm × 76 mm✓ lengths of timber used to support metal roofing sheets.✓ (4)

2.11
- Spreader bars
- Belt slings (2)

2.12 Washers prevent the head of the bolt from pulling through the timber. (1)
[40]

QUESTION 3

3.1
3.1.1 D
3.1.2 E
3.1.3 A
3.1.4 B
3.1.5 C

(5 × 1) (5)

3.2
- Set out the truss template.
- Lay the timber truss members on the template and cut them to size.
- Assemble the truss using nails and other timber connectors. (3)

3.3
3.3.1 Any place on the truss✓ where two or more members are joined.✓

3.3.2 The angle between the top chord and the tie beam✓ expressed in degrees.✓

3.3.3 The truss end joint✓ where the top and bottom chords connect.✓

3.3.4 The top of the truss✓ where the two top chords or rafters meet.✓

(4 × 2) (8)

3.4 Rise = half span × tangent of pitch✓
= 2,25 m × 0,577✓
= 1,299 m →✓ (3)

3.5 The rafter. (1)

[20]

QUESTION 4:

4.1
- Wear the necessary PPE (Personal Protective Equipment).
- Wear a safety harness.
- Do not use faulty equipment.
- Do not work on a roof in windy or wet weather. (2)

4.2
- At the end of the roof.
- In the centre of the roof. (2)

4.3
- It should be kept straight and vertical when carried.
- It should not be allowed to sag or bend.
- Make sure that they do not bend sideways when you slide them over the walls and when you position them upright on the wall. (3)

4.4 Covering the roof✓ with a specific roofing material.✓ (2)

4.5 Two workers check alignment at the eaves✓ and one worker at the apex.✓ (2)

4.6
- Wedges
- Packing
- Caulking mortar (3)

4.7
4.7.1 True
4.7.2 False
4.7.3 False
4.7.4 True
4.7.5 False
4.7.6 False

(6 × 1) (6)
[20]

TOTAL: 100

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